

Using the reflection principle

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Consider a random walk $\{X_n: n \geq 0\}$ with parameter p . Assume $X_0 = 0$, so that $\mathbb{P}(X_0 = 0) = 1$. The (n, X_n) -plane representation is employed, where the first n steps, $X_0 = 0, X_1 = r_1, \dots, X_n = r_n$, correspond to the path $(0, 0) \rightarrow (1, r_1) \rightarrow \dots \rightarrow (n, r_n)$ in the (n, X_n) -plane.

First visit to a negative coordinate

Let T denote the random variable representing the time of the first visit to a negative coordinate. Specifically, $T = \min\{n \geq 1: X_n = -1\}$. Through the application of generating functions, it has been shown that $\mathbb{P}(T = n) = 0$ for even n , and for $n = 2k - 1$,

$$\mathbb{P}(T = 2k - 1) = \frac{1}{2k - 1} \binom{2k - 1}{k} p^{k-1} q^k, \quad k \geq 1. \quad (1)$$

This section provides a proof of the above formula using a counting argument based on the *reflection principle*.

When $T = 2k - 1$, the initial $2k - 1$ steps of the random walk define a path $\mathcal{P}_g$ that starts at $(0, 0)$, ends at $(2k - 1, -1)$, and first crosses the horizontal n -axis at $(2k - 2, 0)$. Such a path is termed a *good path*. Figure 1 illustrates a good path for $T = 15$.

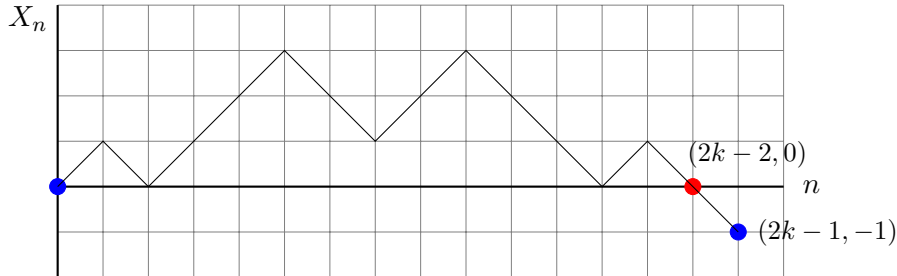


Figure 1: A good path for $T = 15$.

Each path from $(0, 0)$ to $(2k - 1, -1)$ consists of $k - 1$ upward steps $\nearrow$ and k downward steps $\searrow$. Consequently, every trajectory of the random walk with $X_{2k-1} = -1$ occurs with probability $p^{k-1}q^k$. Therefore, if C denotes the number of good paths from $(0, 0)$ to $(2k - 1, -1)$, then,

$$\mathbb{P}(T = 2k - 1) = C p^{k-1} q^k, \quad k \geq 1.$$

Hence, to establish expression (1), it is sufficient to show that

$$C = \frac{1}{2k - 1} \binom{2k - 1}{k}.$$

The coefficient C can also be interpreted as the number of paths from the point $(0, 0)$ to $(2k - 2, 0)$ that do not cross the horizontal n -axis. Moreover, a path from $(0, 0)$ to $(2k - 1, -1)$ that is not a good path can be seen as a *bad path* $\mathcal{P}_b$ from $(0, 0)$ to $(2k - 2, 0)$ that *crosses* the horizontal axis at

some point $(r, 0)$ before reaching $(2k - 2, 0)$. Figure 2 illustrates such a bad path. Therefore, C is given by the difference $K - B$, where K denotes the total number of paths from $(0, 0)$ to $(2k - 2, 0)$ and B is the number of bad paths $\mathcal{P}_b$. Since a path from $(0, 0)$ to $(2k - 2, 0)$ consists of $k - 1$ steps $\nearrow$ and $k - 1$ steps $\searrow$, K is the number of ways to choose $k - 1$ steps $\nearrow$ from a total of $2k - 2$ steps. Thus,

$$K = \binom{2k - 2}{k - 1}.$$

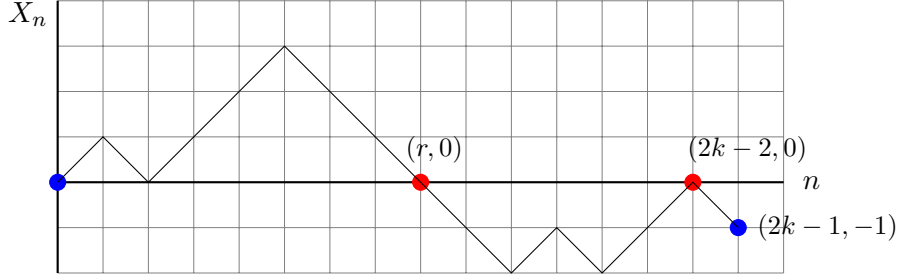


Figure 2: A bad path.

To determine the value of B , the reflection principle can be applied, as illustrated in Figure 3. Let $\mathcal{P}_b$ denote a bad path from $(0, 0)$ to $(2k - 2, 0)$, and let $\mathcal{P}_R$ denote the subpath of $\mathcal{P}_b$ that begins

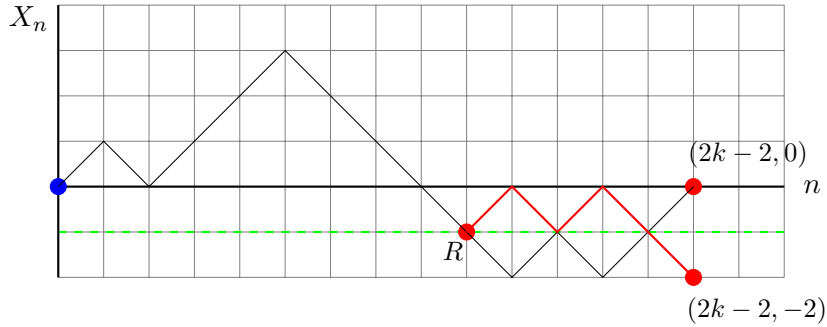


Figure 3: A reflected “bad” path.

at the point R , corresponding to the first visit of the random walk to the coordinate -1 . If $\mathcal{P}_R$ is reflected around the dashed green line $y = -1$, then $\mathcal{P}_b$ is transformed into a path from $(0, 0)$ to $(2k - 2, -2)$. Conversely, any path from $(0, 0)$ to $(2k - 2, -2)$ can be mapped back to a bad path from $(0, 0)$ to $(2k - 2, 0)$ via the inverse reflection process. Thus, this reflection process establishes a bijection between the set of bad paths $\mathcal{P}_b$ and the set of all paths from $(0, 0)$ to $(2k - 2, -2)$. Consequently,

$$B = \binom{2k - 2}{k - 2},$$

since a path from $(0, 0)$ to $(2k - 2, -2)$ consists of $k - 2$ steps $\nearrow$ and k steps $\searrow$.

Therefore,

$$\begin{aligned} C = K - B &= \binom{2k - 2}{k - 1} - \binom{2k - 2}{k - 2} = \frac{(2k - 2)!}{(k - 1)!(k - 1)!} - \frac{(2k - 2)!}{(k - 2)!k!} \\ &= \frac{(2k - 2)!(k - (k - 1))}{(k - 1)!k!} = \frac{1}{2k - 1} \cdot \frac{(2k - 1)!}{(k - 1)!k!} = \frac{1}{2k - 1} \binom{2k - 1}{k}, \end{aligned}$$

as we wanted to prove.

First return to the origin.

Consider a random walk $\{X_n: n \geq 0\}$ with parameter p , and define $Z = \min\{n \geq 1: X_n = 0\}$ as the time of the first return to zero. The reflection principle is applied in this section to establish that

$$\mathbb{P}(Z = 2k) = \frac{1}{2k-1} \binom{2k}{k} p^k q^k, \quad k \geq 1. \quad (2)$$

If $X_{2k} = 0$, the initial $2k$ steps of the trajectory of $\{X_n: n \geq 0\}$ define, in the (n, X_n) -plane, a path that begins at $(0, 0)$ and terminates at $(2k, 0)$. The probability of such a path is $p^k q^k$. Therefore, $\mathbb{P}(Z = 2k)$ can be expressed as $C p^k q^k$, where C represents the number of paths from $(0, 0)$ to $(2k, 0)$ that do not intersect the horizontal n -axis at any point $(n, 0)$ for $1 \leq n \leq 2k - 1$. To establish the formula in equation (2), it is sufficient to show that

$$C = \frac{1}{2k-1} \binom{2k}{k}.$$

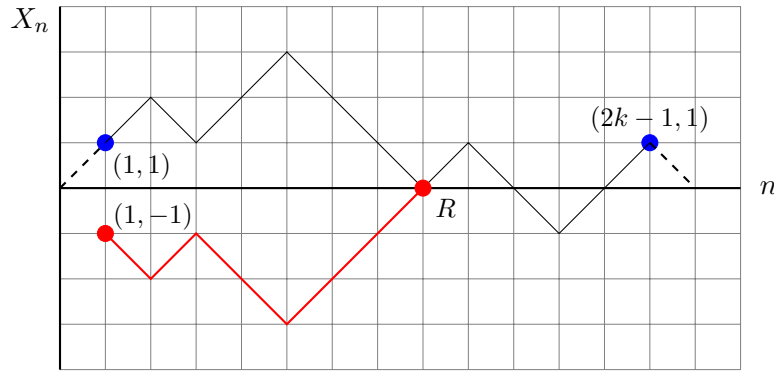


Figure 4: A reflected “bad” path.

We observe that $C = C_1 + C_2$, where C_1 represents the number of paths from $(1, 1)$ to $(2k-1, 1)$ that do not touch the n -axis, and, symmetrically, $C_2 = C_1$ represents the number of paths from $(1, -1)$ to $(2k-1, -1)$ that avoid the n -axis. Therefore, we can express C as $C = 2C_1$. Furthermore,

$$C_1 = \binom{2k-2}{k-1} - C',$$

where C' denotes the number of paths from $(1, 1)$ to $(2k - 1, 1)$ that touch or cross the n -axis. By the reflection principle, C' is also equal to the total number of paths from $(1, -1)$ to $(2k - 1, 1)$. (See Figure 4.) This quantity is given by

$$C' = \binom{2k-2}{k},$$

since a path from $(1, -1)$ to $(2k - 1, 1)$ consists of k upward steps $\nearrow$ and $k - 2$ downward steps $\searrow$.

We conclude that

$$\begin{aligned} C = 2C_1 &= 2 \left(\binom{2k-2}{k-1} - \binom{2k-2}{k} \right) = 2 \left(\frac{(2k-2)!}{(k-1)!(k-1)!} - \frac{(2k-2)!}{k!(k-2)!} \right) \\ &= 2 \frac{(2k-2)!}{k!(k-1)!} = \frac{2}{2k-1} \binom{2k-1}{k} = \frac{1}{2k-1} \left(\binom{2k-1}{k} + \binom{2k-1}{k-1} \right) \\ &= \frac{1}{2k-1} \binom{2k}{k}, \end{aligned}$$

as we wanted to prove.

Maximum in n steps.

Consider a symmetric random walk $\{X_n: n \geq 0\}$, and define $M_n = \max(X_0, X_1, \dots, X_n)$ as the maximum position attained during the first n steps. In this section, we will demonstrate that for $0 \leq r \leq n$, the probability $\mathbb{P}(M_n = r)$ equals $\mathbb{P}(X_n = r)$ when n and r share the same parity (both even or both odd), whereas if n and r have different parity, then $\mathbb{P}(M_n = r) = \mathbb{P}(X_n = r + 1)$.

We begin by applying the reflection principle to demonstrate that

$$\mathbb{P}(X_n = k, M_n \geq r) = \mathbb{P}(X_n = 2r - k), \quad 0 \leq r \leq n, \quad -n \leq k \leq r. \quad (3)$$

We assume that n and k have the same parity, since the probabilities in equality (3) are otherwise zero. In the (n, X_n) -plane representation of the random walk, any path ending at (n, k) with $M_n \geq r$ must cross the horizontal line $y = r$. Considering the reflection R shown in Figure 5, the number of such paths is equal to the number of paths from $(0, 2r)$ to (n, k) . This is also equivalent to the number of paths from $(0, 0)$ to $(n, k - 2r)$, as illustrated in Figure 5. Therefore, since all paths with n steps are equally probable, the fraction of paths ending at (n, k) for which $M_n \geq r$ is given by $\mathbb{P}(X_n = k, M_n \geq r) = \mathbb{P}(X_n = k - 2r)$. By symmetry, this is also equal to $\mathbb{P}(X_n = 2r - k)$. This demonstrates equality (3).

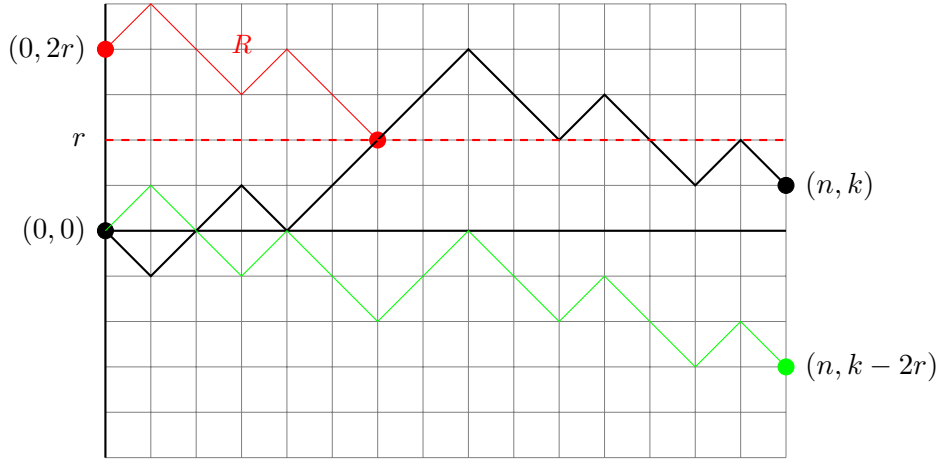


Figure 5: Reflection and translation of a $(0, 0) \rightarrow (n, k)$ path such that $M_n \geq r$.

Now, for each k we have

$$\begin{aligned} P(X_n = k, M_n = r) &= \mathbb{P}(X_n = k, M_n \geq r) - \mathbb{P}(X_n = k, M_n \geq r + 1) \\ &= \mathbb{P}(X_n = 2r - k) - \mathbb{P}(X_n = 2(r + 1) - k), \end{aligned}$$

and adding for all $k \leq r$, we get

$$\mathbb{P}(M_n = r) = \sum_{k \leq r} \mathbb{P}(X_n = k, M_n = r) = \sum_{k=-n}^r (\mathbb{P}(X_n = 2r - k) - \mathbb{P}(X_n = 2(r + 1) - k)).$$

This is a telescoping sum yielding

$$\begin{aligned} \mathbb{P}(M_n = r) &= -\mathbb{P}(X_n = 2r + 2 + n) - \mathbb{P}(X_n = 2r + 1 + n) + \mathbb{P}(X_n = r) + \mathbb{P}(X_n = r + 1) \\ &= \mathbb{P}(X_n = r) + \mathbb{P}(X_n = r + 1). \end{aligned}$$

The last equality holds because the terms $\mathbb{P}(X_n = 2r + 2 + n)$ and $\mathbb{P}(X_n = 2r + 1 + n)$ are both zero for any value of r . Furthermore, among the remaining two terms, one is zero due to parity considerations. Therefore, $\mathbb{P}(M_n = r) = \mathbb{P}(X_n = r)$ if n and r have the same parity; otherwise, $\mathbb{P}(M_n = r) = \mathbb{P}(X_n = r + 1)$, as required.